

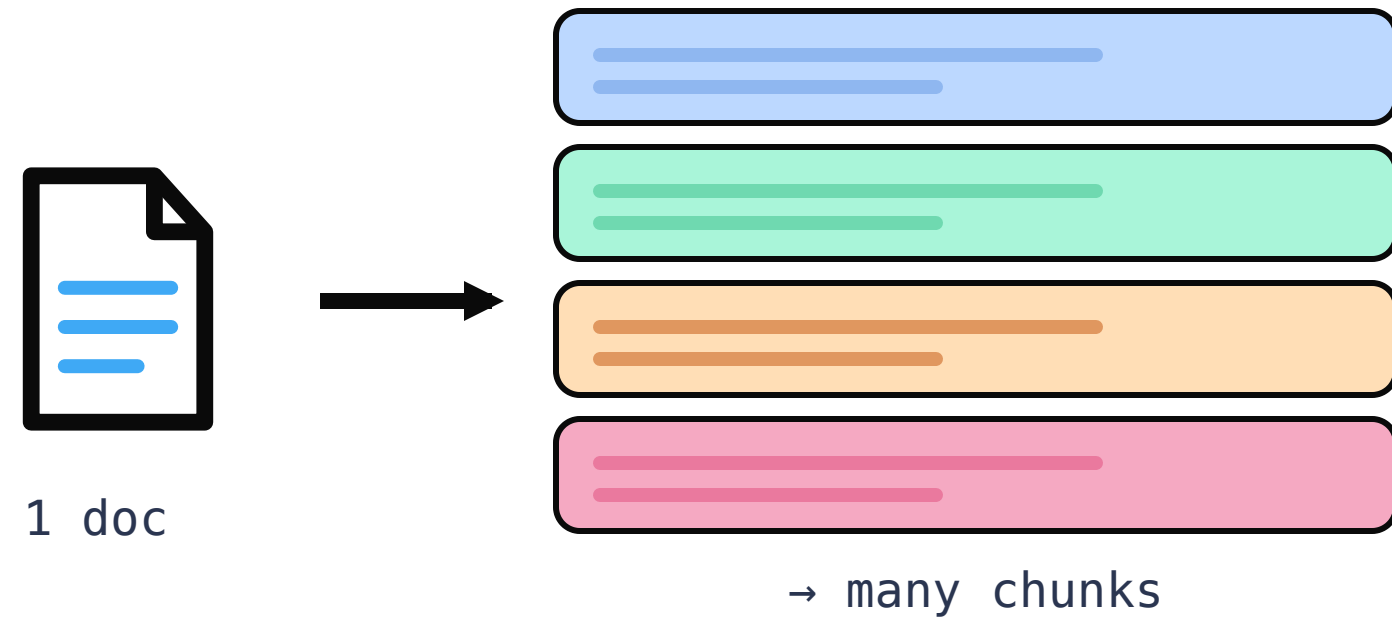
// PANGAEA LABS • SYSTEM DESIGN

How **RAG** Works

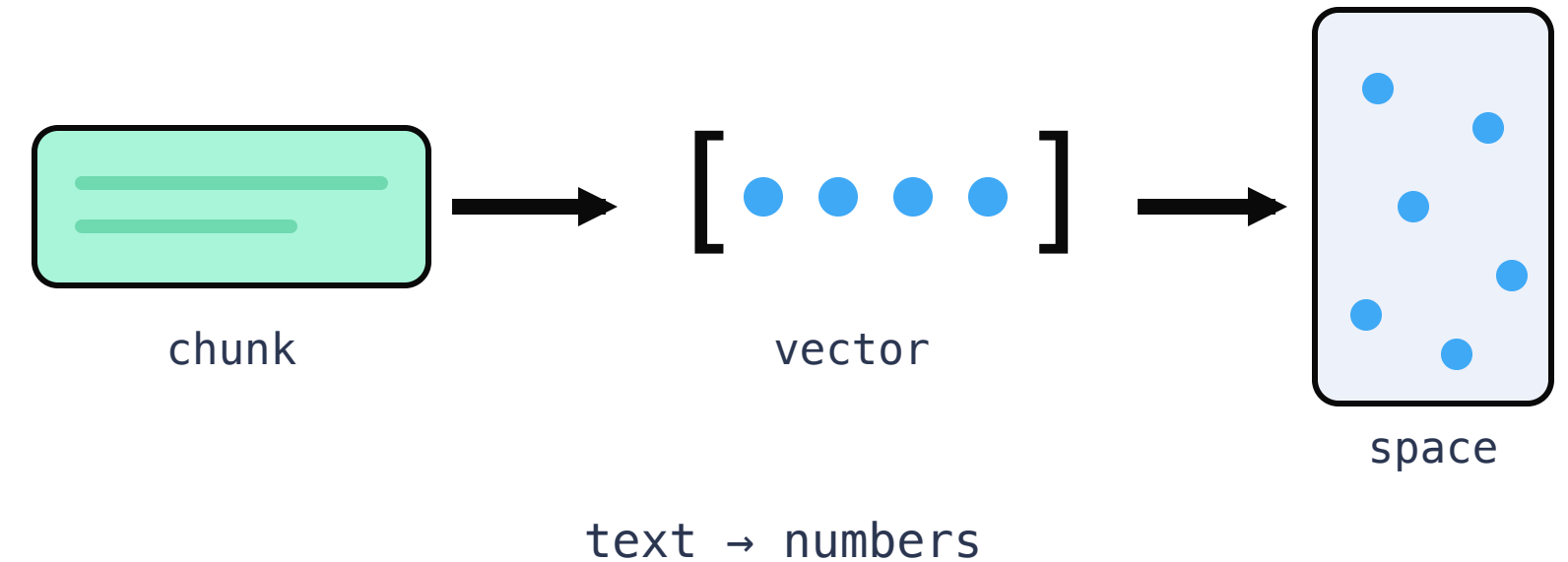
A concept-card walkthrough of retrieval-augmented generation — every concept is one hand-written SVG.

RAG, end to end

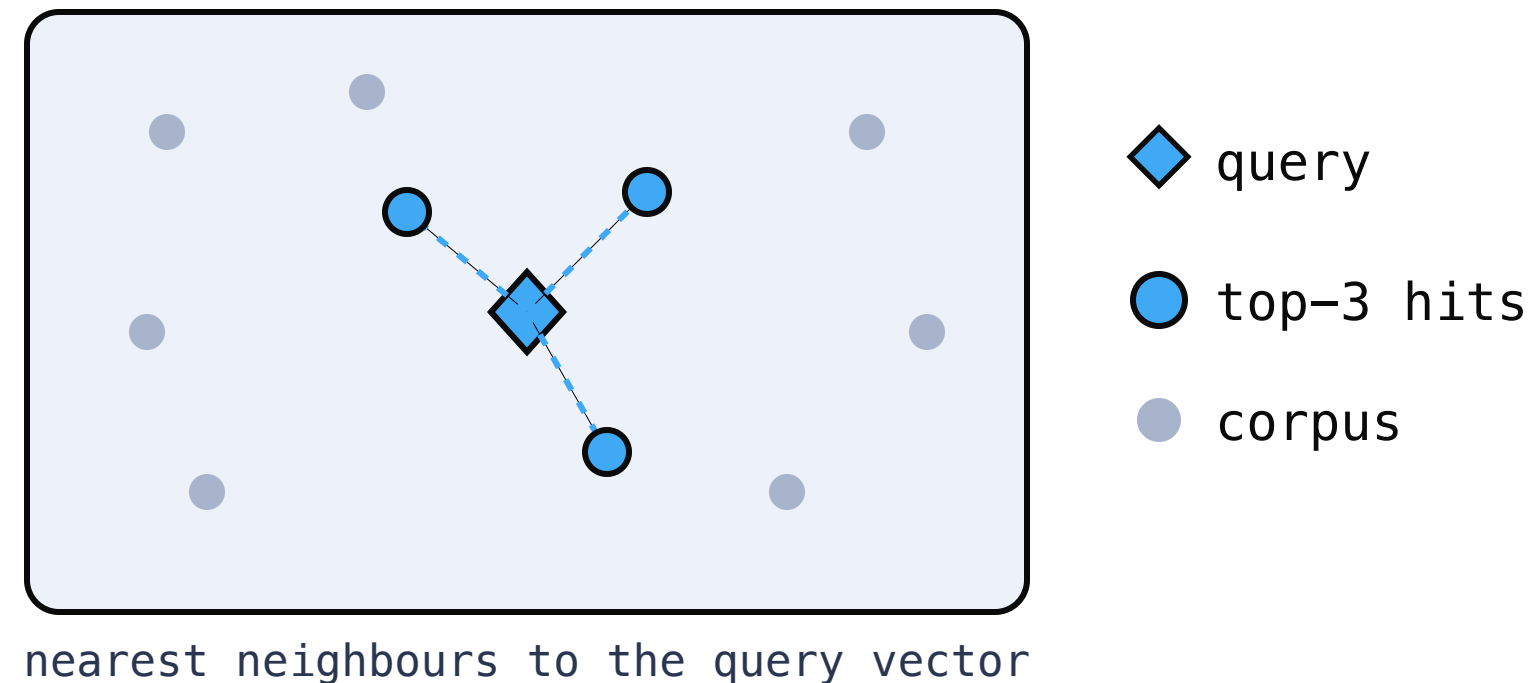
1 · Chunk



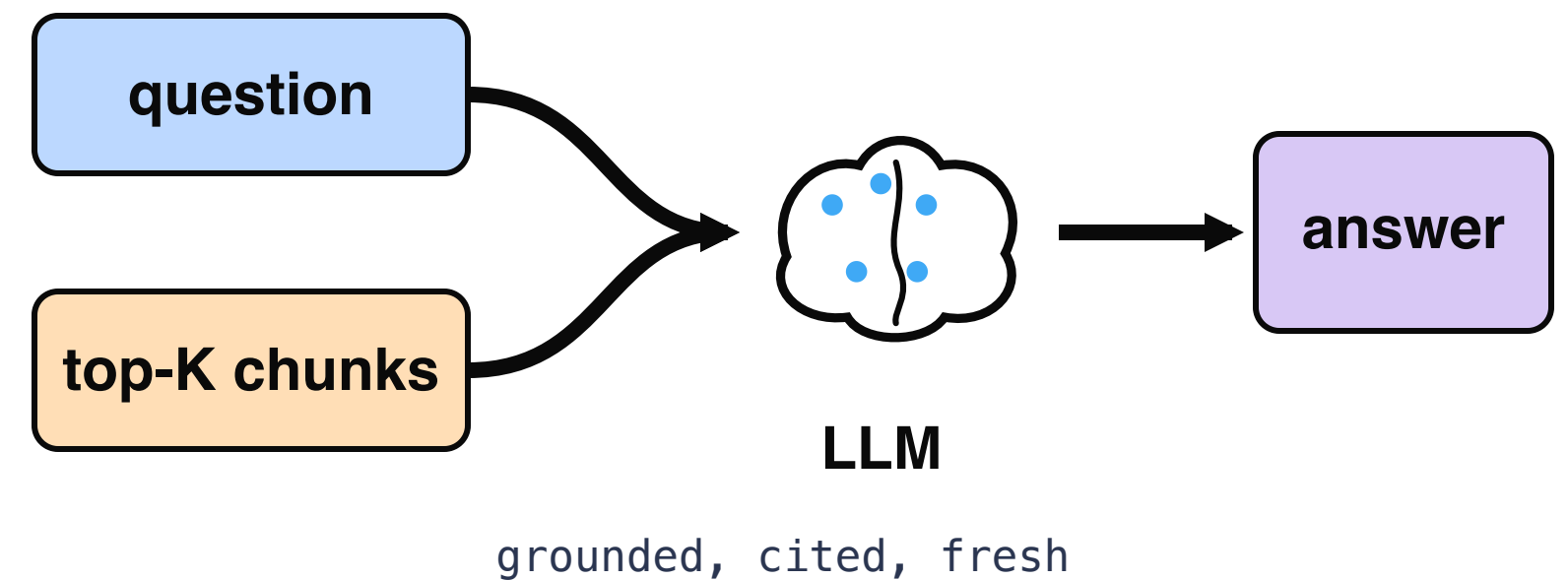
2 · Embed



3 · Retrieve top-K



4 · Augment & generate

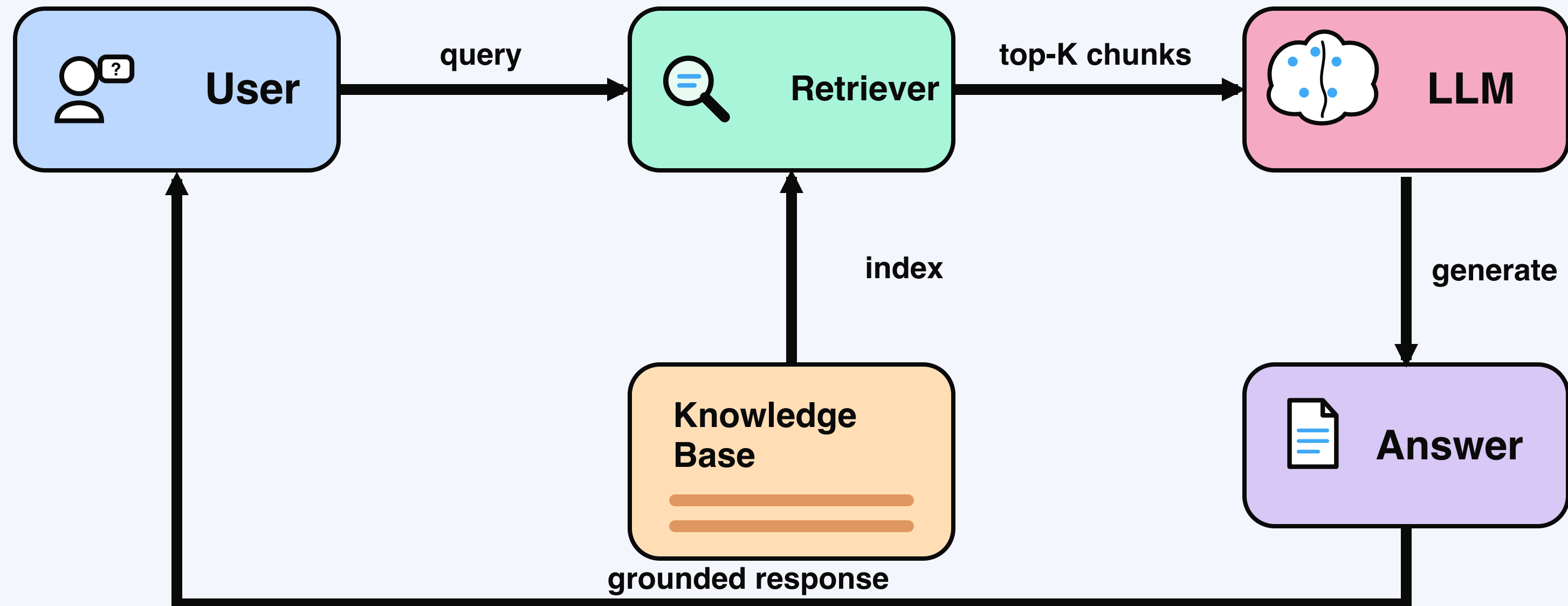


Without RAG, it guesses

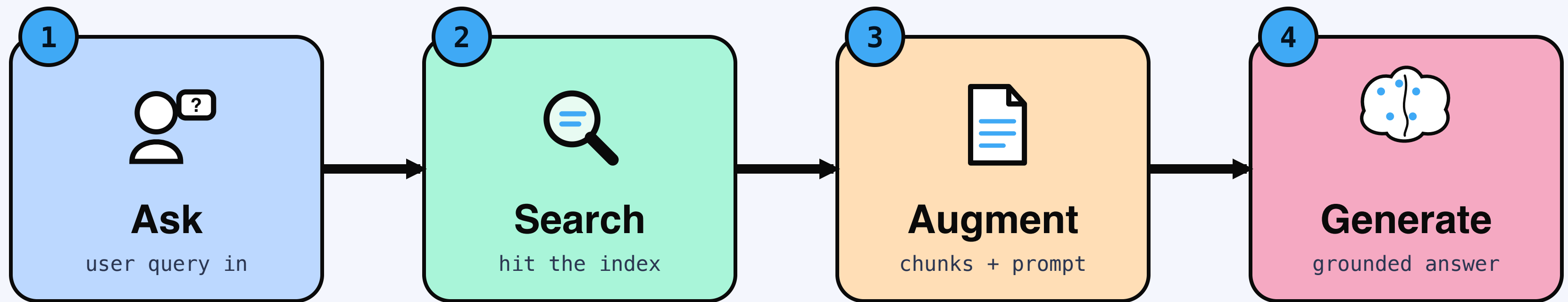


It answers from memory — and **bluffs** when it doesn't know.

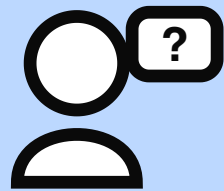
The RAG architecture



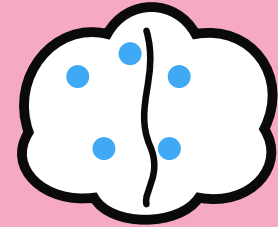
The RAG pipeline



Fetch first — **then** generate.



User



LLM / brain



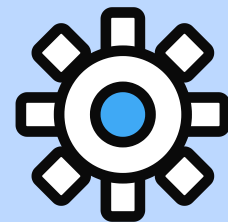
Search



Database



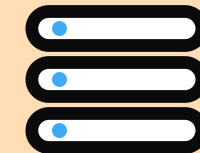
Document



Process / gear



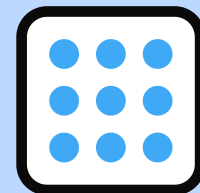
Cache



Queue



API / cloud



Embeddings

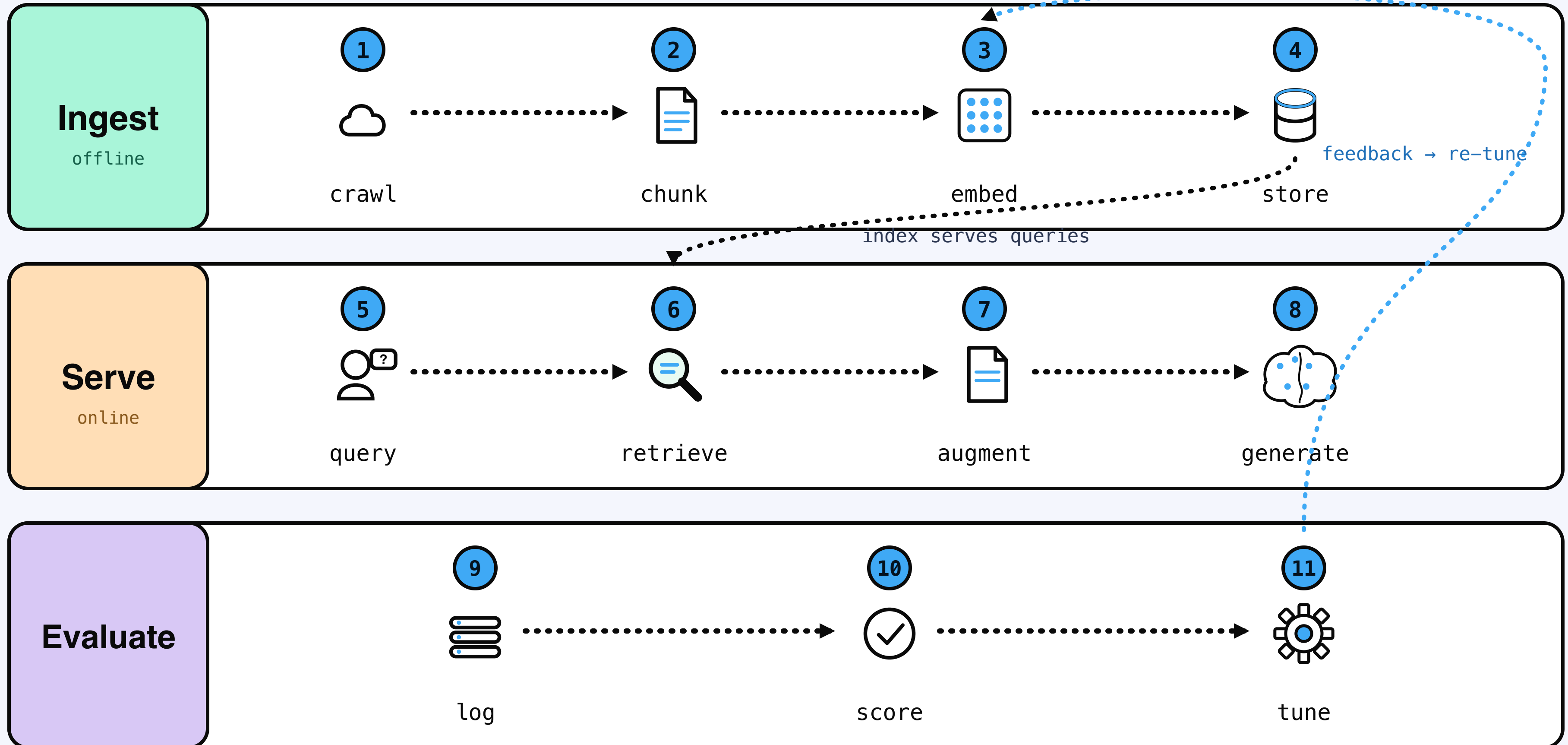


Auth / lock



Verdict

The RAG lifecycle



// REMEMBER

Fine-tuning edits the **brain. RAG
edits the **desk**.**

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Build with RAG

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